

Fig. 3: A single-side, actual-size PCB layout for microcontroller-based solar charger

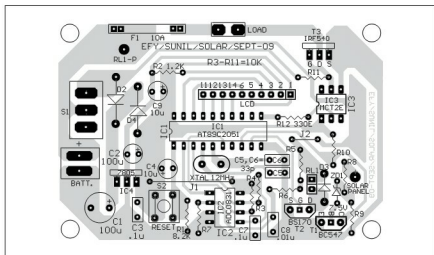


Fig. 4: Component layout for the PCB

given below.

Microcontroller. Microcontroller AT89C2051 is the heart of the circuit. It is a low-voltage, high-performance, 8-bit microcontroller that features 2 kB of Flash, 128 bytes of RAM, 15 input/output (I/O) lines, two 16-bit timers/counters, a five-vector two-level interrupt architecture, a full-duplex serial port, a precision analogue comparator, on-chip oscillator and clock circuitry. A 12MHz crystal is used for providing the basic clock frequency. All I/O pins are reset to '1' as soon as RST pin goes high. Holding RST pin high for two machine cycles, while the oscillator is running, resets the device. Power-on reset is derived from resistor R1 and

capacitor C4. Switch S2 is used for manual reset.

Serial ADC. The microcontroller monitors the battery voltage with the help of an analogue-to-digital converter. The ADC0831 is an 8-bit successive approximation analogue-to-digital converter with a serial I/O and very low conversion time of typically 32 μ s. The differential analogue voltage input allows increase of the common-mode rejection and offsetting of the analogue zero input voltage. In addition, the voltage reference input can be adjusted to allow encoding of any smaller analogue voltage span to the full eight bits of resolution. It is available in an 8-pin PDIP package and

PARTS LIST

Semiconductors:

IC1	- AT89C2051 microcontroller
IC2	- ADC0831 analogue-to-digital converter
IC3	- MCT2E optocoupler
IC4	- 7805, 5V regulator
T1	- BC547 npn transistor
T2	- BS170 n-channel MOSFET
T3	- IRF540 n-channel MOSFET
D1	- 6A4 rectifier diode
D2-D4	- 1N4007 rectifier diode
ZD1	- 7.5V zener diode

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon):

R1	- 8.2-kilo-ohm
R2	- 1.2-kilo-ohm
R3, R4, R6-R11	- 10-kilo-ohm
R5	- 20-kilo-ohm
R12	- 330-ohm

Capacitors:

C1	- 100 μ F, 63V electrolytic
C2	- 100 μ F, 16V electrolytic
C3, C7	- 0.1 μ F ceramic disk
C4, C9	- 10 μ F, 16V electrolytic
C5, C6	- 33pF ceramic disk
C8	- 0.01 μ F ceramic disk

Miscellaneous:

S1	- On/off switch
S2	- Push-to-on switch
RL1	- 12V, 1C/O relay
X ₁	- 12MHz crystal
LCD	- 16 $\times$ 2 line display
Solar panel	- 10-40W
	- 10A fuse

can be interfaced to the microcontroller with only three wires.

LCD module. The system status and battery voltage are displayed on an LCD based on HD44780 controller. The backlight feature of the LCD makes it readable even in low light conditions. The LCD is used here in 4-bit mode to save the microcontroller's port pins. Usually the 8-bit mode of interfacing with a microcontroller requires eleven pins, but in 4-bit mode the LCD can be interfaced to the microcontroller using only seven pins.

Solar panel. The solar panel used here is meant to charge a 12V battery and the wattage can range from 10 to 40 watts. The peak unloaded voltage output of the solar panel will be around 19 volts. Higher-wattage panels can be used with some modifications to the controller unit.

Rechargeable battery. The solar energy is converted into electrical energy and stored in a 12V lead-acid battery. The ampere-hour capacity ranges from